Lunar Logistics Rover (LLR)

A First-Pass Concept Study for a Canadian Lunar Utility Vehicle

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Executive Summary

The Lunar Logistics Rover (LLR) is a concept design that answers the need for a future large, multipurpose crewed lunar rover. Recently, the Canadian Space Agency (CSA) has expressed interest in filling this demand via the Lunar Utility Vehicle (LUV) [1] program, calling on Canadian aerospace companies to develop proposals. This report attempts to outline and solve some of the challenges of this development, and propose a solution.

The rover is designed for the lunar south pole region, the location of NASA's planned lunar surface base. The LLR is sized for a 3,500 kg payload in modular cargo, including a crew of two astronauts for up to 8 hours at a time, and is designed for reliability and survival on the lunar surface. Table 1 gives the LLR's key figures.

Table 1: LLR at a glance.

Loaded mass	5,250 kg (3,500 kg payload + 1,750 kg targeted dry mass)
Wheels	Six 1.2 m wheels, 90 kg each (estimated), 5 cm grousers
Rolling resistance	0.36 of wheel load (Bekker analysis using NASA soil data)
Steepest slope	20° required, but this report was unable to close this while accounting for slip. The grip 'ceiling' is 20.4° with grousers, but this actually can't be reached at any real slip value, meaning a traction test is needed to verify
Day survival	5.9 m^2 sky-facing radiator dissipates drive unit heat
Night survival	255 W (estimated) heat load, met by MMRTG heat (1,790 W at 14 years) through a pumped fluid loop system
Wheel flexure	Factor of safety 1.68 at a 3,402 N worst-case load, 5.3 kg each

Four main studies were used to derive these numbers. The mobility section performs a Bekker regolith analysis to test whether a large rover can meet them. It can drive on regolith, but only with a significant penalty of about 36% of the rover's weight-force acting counter to the wheels just to move it on level ground. Additionally, the 20-degree slope requirement requires physical testing, as the model predicts it is not met by the current design, instead demanding unfeasibly large grousers. Drive unit selection is performed after this, to select motors and gearboxes for the drivetrain. These numbers feed into a thermal analysis that considers what the LLR needs to do to survive a harsh lunar south-pole night, as well as how to keep itself cool during daytime operations. The study shows that a pumped fluid loop paired with a Multi-Mission Radioisotope Thermoelectric Generator (MMRTG) safely keeps the LLR warm through the night with energy to spare. However, the radiator that cools the drive units should lie flat and face the sky, which strays from traditional design norms of non-lunar rovers. Finally, an FEA wheel study was conducted to demonstrate the feasibility of large-scale lunar wheels. Previous studies have shown that Apollo-era wheels do not scale well, but this report proposes a new design combining CSA and Mars-rover wheel design philosophies to counter the challenge of lunar driving.

Generally outside the scope of this report are some important aspects: communications architecture, cargo handling, and modular payload support are left off due to the short time span of the project. Additionally, this report is only one answer to the ongoing question of how to move things on the Moon. As this question is constantly changing, aspects of this proposal can and will change, as it is a work-in-progress.

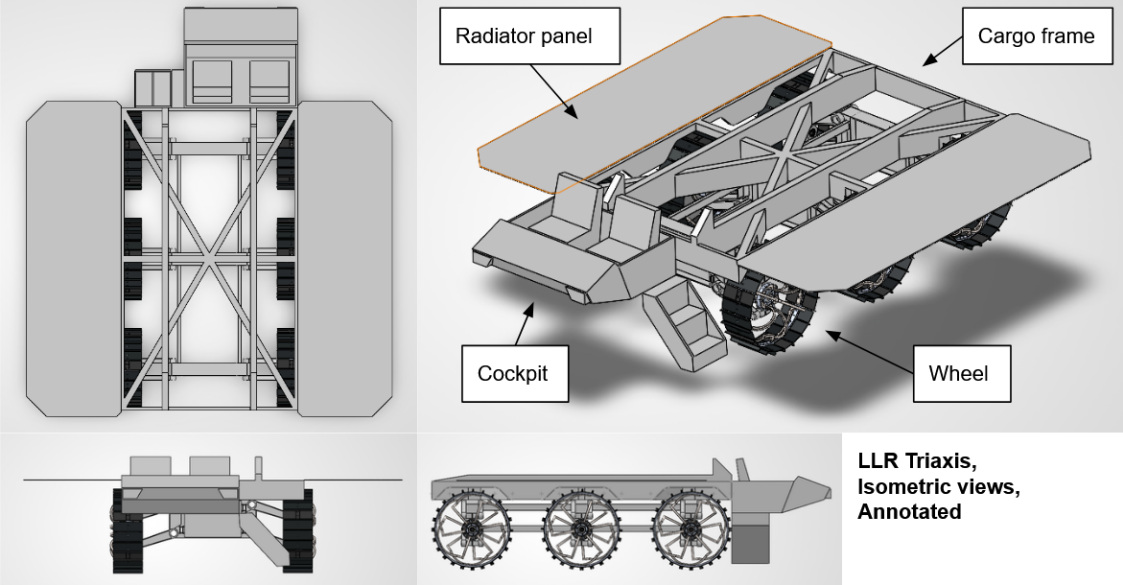


Figure 1: Preliminary CAD concept of the LLR, showing main components such as wheels, cockpit, and side-mounted radiator fins.

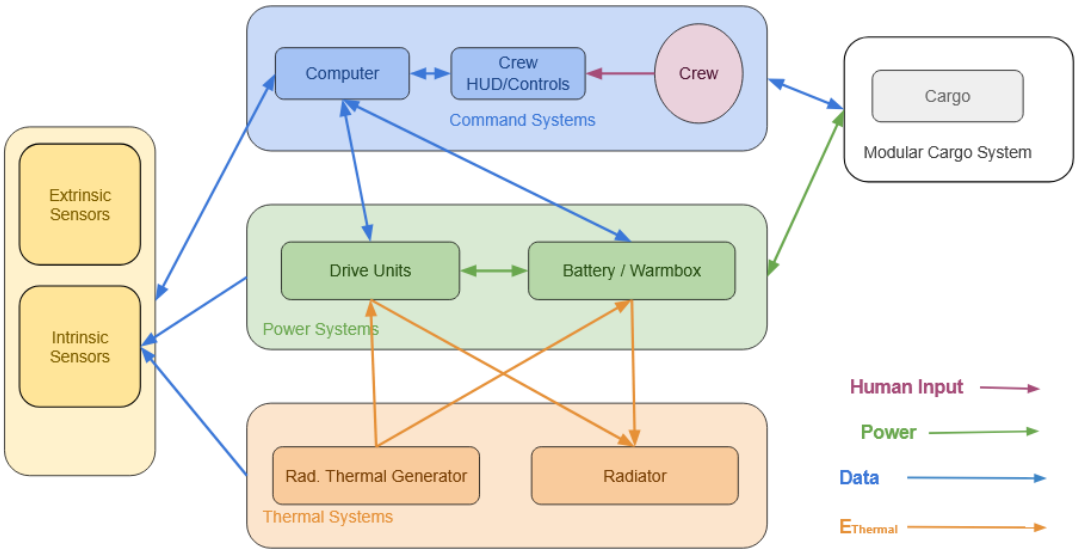


Figure 2: LLR system block diagram: command, power, thermal, sensing and cargo systems, with human input, power, data and thermal flows.

1 Mission Context (CONOPS)

With the Artemis program, returning to the Moon has become a central goal of the present-day space sector. The Canadian space sector once again has a critical role to play, developing a modern cargo transport vehicle for the lunar surface: the Lunar Utility Vehicle (LUV) [1]. The rover outlined in this report is an evolution of that concept, and builds off of assumed payload, size and

function requirements.

The LLR is designed to be both survivable and versatile. For this reason, it is able to fit within the inner fairing of both potential lunar landing craft (the SpaceX 'Starship' (8.0 m) [3] and Blue Origin 'New Glenn' / 'Blue Moon Mk1' (7.0 m) [4]). By folding forward of the front axle, the LLR keeps its cargo deck in one piece for structural strength while maintaining a 5.75 m diameter envelope.

After landing, the LLR unfolds itself, deboards the landing craft, is equipped with the necessary scientific or cargo payload, and is deployed in one of three operating modes (Figure 3).

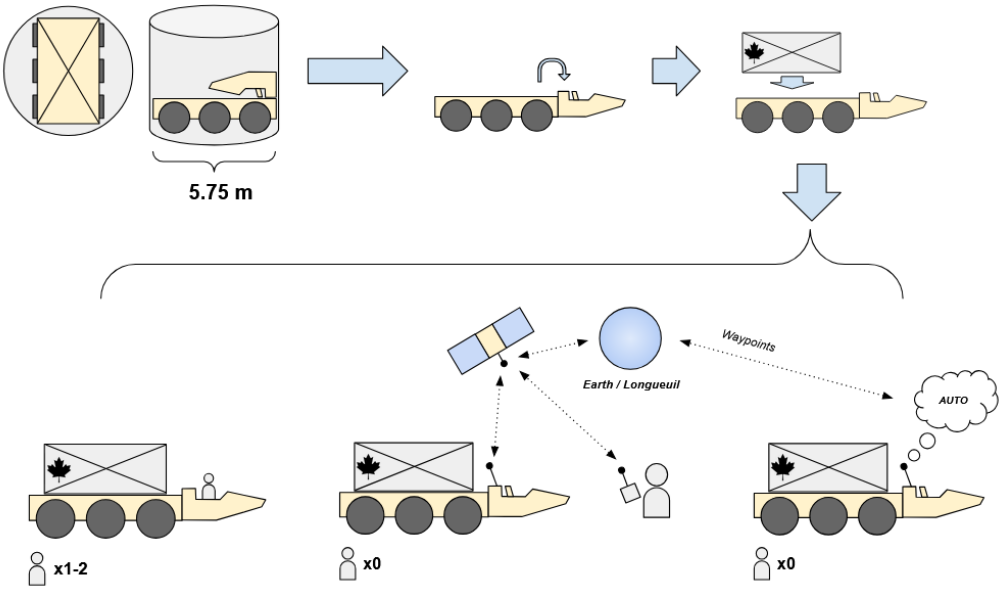


Figure 3: LLR conceptual operations plan, showcasing deployment, and three possible modes of operation.

Crewed mode allows the LLR to transport itself and up to two Artemis astronauts to conduct transport, scientific, and survey missions for up to 8 hours, the limit of Artemis Axiom AxEMU suit life support. Remote operation allows the LLR to be operated by crew members on the Moon, or mission control on Earth to complete longer, less complex missions or to minimize crew risk outside of the lunar habitat. However, transmission delay could realistically limit LLR speeds. Autonomous waypoint following allows the LLR to receive positional instructions from Earth or the lunar habitat, and allow itself to make the best decisions to get there without crew input, thus bypassing any transmission speed delays. Additionally, in case of lost contact with the LLR during remote operation, this would allow the rover to return to a point where it can communicate, or the lunar base itself.

During missions, the LLR is built to accommodate modular payloads to transport lunar cargo, complete scientific or sensing tasks, transport crew members to and from launch sites or aid in lunar construction tasks. Additionally, this modular ability would allow the LLR to extend its ability in the future by adding more battery capacity, solar charging, or life support capability.

After a mission, the LLR should return to the lunar habitat to charge and be maintained prior to

another mission. However, the LLR is designed to keep itself alive through a 150 h dormant period via RTG heating, even if not connected to a power grid and without battery assistance.

The LLR is designed for the CSA's 10 year lifespan, and limits modes of failure through a simple, rugged design. However, the lunar south pole region is harsh, facing extremely low temperatures, potential lack of sunlight, rough terrain and dangerous permanently shadowed regions (PSRs). For this reason, extensive safety rules will have to be adhered to, including operating procedures regarding entering PSRs, maximum distance from base, minimum battery levels et cetera. With proper operation, the LLR should be a rugged and reliable vehicle that helps build a permanent human base on the Moon safely, and can support whatever work comes after that.

2 Requirements and Baseline

The design of the LLR is recorded in a 'decision register' (Appendix A). This register records the input requirements, and the decisions derived from those requirements or from the studies outlined below. Most input values come from recent CSA and NASA literature on the Artemis missions, the VIPER rover and the LUV project. Missing numbers, such as speed, were filled in with judgement and the specifications of the last crewed lunar rover, the Apollo Lunar Roving Vehicle (LRV).

2.1 Base Requirements

The full list of requirements can be found in Appendix A, but only a few of them actually shape most of the design, and these are outlined here.

The most important of these is payload. The LLR must carry 3,500 kg, crew included (Payload-1), based on industry reporting on the LUV [2]. Together with a targeted dry mass of 1,750 kg (Payload-2), this gives a 5,250 kg loaded mass, which is used for the wheel load, torque and power calculations throughout this report. The dry mass target is a 2:1 payload ratio, chosen with the Apollo LRV's 2.33:1 in mind.

For mobility, the LLR must climb a 20° slope at full load (Mobility-4), taken from NASA's LTV requirements [5]. This sets how much grip the wheels need, and led to the 5 cm grousers in Section 3. The LLR must also climb 15° with one drive unit failed (Mobility-5), and climb a 30.5 cm obstacle like the Apollo LRV (Mobility-2) [6], and the drive units are sized around these two cases. Top speed is 4 km/h on level ground (Mobility-1), which sets the 120:1 gear ratio and the heat the radiator has to get rid of.

For survival, the LLR must survive 150 h dormant (Environment-2) and last at least 10 years (Environment-4) [5, 1], which is why the MMRTG and pumped fluid loop were chosen. The RTG is checked at 14 years, because the MMRTG's published end-of-life figure is at 17 years, including 3 years of fuelled storage before launch [13].

2.2 How the Numbers Flow Down

The payload and dry mass give the loaded mass, and that sets the load on every wheel:

$$M = 3500 + 1750 = 5{,}250\,\text{kg} \tag{1}$$

$$W = 5250 \times 1.62 = 8{,}505\,\text{N} \tag{2}$$

$$N_w = 8505 \div 6 = 1{,}417.5\,\text{N per wheel} \tag{3}$$

That wheel load is then used in the Bekker soil model (Section 3), which gives a rolling resistance of $C_{rr} = 0.357$. From there, each part of the design is sized by a different case, as outlined below.

The hardest continuous torque case is Mobility-5: climbing 15° with one drive unit failed, so five wheels do the work of six:

$$F = W\,(C_{rr}\cos\theta + \sin\theta) = 8505 \times (0.357 \times 0.966 + 0.259) = 5{,}136\,\text{N} \tag{4}$$

$$T = \frac{5136}{5} \times 0.60 = 616\,\text{N m per working wheel} \tag{5}$$

The obstacle (Mobility-2) sets the peak torque at 741 N m. These two numbers are then checked against the gearbox's 688 N m rating and 1,404 N m repeated peak to size the drive unit (Section 4).

However, the radiator is sized by a different case entirely. The drive units actually make the most heat while cruising at 4 km/h on level ground, at only 304 N m per wheel, because that is where the most power goes through the drivetrain. This case is used to size the radiator in Section 5.8.

Lastly, the 150 h night and the 10-year life set the RTG, which keeps the vehicle warm through the fluid loop (Section 5.2).

2.3 Where the Design Does Not Yet Meet Its Requirements

This section outlines where the LLR does not yet meet its requirements, or where this report has not been able to show that it does yet. The largest of these is grip on slopes, but there are also several numbers that have not been budgeted or tested yet, which are covered afterwards.

The first problem is the 20° slope requirement (Mobility-4). To climb 20°, each wheel needs grip equal to the rolling resistance plus the slope (Section 3.11):

$$\mu_{req} = C_{rr} + \tan 20^\circ = 0.357 + 0.364 = 0.721 \tag{6}$$

With 5 cm grousers, the most grip a wheel can get is $H_{max} = 1{,}034.7$ N (Section 3.13). Taking away the rolling resistance, the steepest slope that grip could ever hold is

$$\tan\theta_{max} = \frac{1034.7}{1417.5} - 0.357 = 0.3729 \quad \Rightarrow \quad \theta_{max} = 20.4^\circ \tag{7}$$

This is only just above 20°, and is only a ceiling, as the soil only gives grip as the wheel slips. Even at 100% slip, the wheel only produces 937.4 N, 84.6 N short of the 1,022 N needed (Section 3.14). The same problem appears in the Mobility-5 case, with one drive unit failed on 15°, where each working wheel must give 0.75 of its load, but at most 0.66 is available.

The Bekker model is likely pessimistic here, since it applies bulldozing to every wheel and keeps the smooth wheel's slip behaviour (Section 3.14). For this reason, the slope requirements are left to a single-wheel traction test in lunar simulant rather than redesigning the wheel around the model.

Apart from grip, there are also some numbers the design depends on that have not been checked yet. There is no mass budget, so the 1,750 kg dry mass (Payload-2), and with it the 3,500 kg payload (Payload-1), are still targets. The wheels are estimated at 540 kg (Section 6), which is almost a third of the dry mass target by themselves. The battery has not been sized either, so the 8 h sortie and the PSR traverse are not yet assessed. Lastly, climbing the 30.5 cm obstacle (Mobility-2) relies on a grouser hooking the rock edge. This works in the calculation (Section 4), but has not been tested in reality.

Table 2: Main requirements not yet met, or not yet shown to be met.

ID	Requirement	Current result	What would close it
Mobility-4	20° slope	Grip ceiling is 20.4° with 5 cm grousers, but the model does not reach it at any slip	South-pole soil data; single-wheel traction test
Mobility-5	15° with one drive failed	Each working wheel must give 0.75 of its load; at most 0.66 is available	As above
Mobility-2	30.5 cm obstacle	Met on paper if a grouser hooks the edge	Single-wheel obstacle test
Payload-1, -2	3,500 kg payload, 1,750 kg dry	No mass budget yet; wheels alone are about 540 kg	Mass budget
Environment-1, -3	2 h PSR, 8 h sortie	Battery not yet sized	Battery sizing

3 Mobility: How Hard Is It to Drive on the Moon?

(Actually, very hard!)

The LLR is far heavier than any vehicle that has driven on the Moon. The first idea was to model its wheels on the Apollo Lunar Roving Vehicle (LRV), for simplicity. However, earlier attempts to scale the LRV wheel up to heavier loads have failed [7]. For this reason, the plausibility of functional 1.2 x 0.45 m wheels was considered from the ground up in this study. This section calculates the force needed to keep a wheel rolling through lunar regolith, and the maximum grip the regolith can enact on the LLR, which allows us to determine how steep a hill the rover can climb.

The method follows the NASA white paper *Terramechanics for LTV: Modeling and Simulation* [8], which applies Bekker's terramechanics model [9]. All soil values, including the soil-strength factors, come from Table 1 of that paper, and the equations are used in the way the paper defines. Steps 1 to 9 start with a smooth, rigid wheel. Steps 10 to 12 add grousers, using the grouser model from the same paper, to determine if they are a worthwhile improvement.

3.1 What Is Rolling Resistance?

The coefficient of rolling resistance, C_{rr}, is the fraction of a wheel's load needed to keep it rolling:

$$F = C_{rr}\, N \tag{8}$$

For example, modern car tyres on smooth concrete have a C_{rr} of about 0.01 [11]: the force needed to push or pull the wheel is about 1% of its load.

On soft regolith, almost all of the resistance comes from two sources, compacting the regolith beneath the wheel, and bulldozing a pile of regolith in front of the wheel. Both are calculated from the NASA method.

3.2 Inputs

Table 3: Vehicle inputs.

Quantity	Symbol	Value	Source
Loaded mass	M	5,250 kg	Payload-1 + Payload-2
Lunar gravity	g	1.62 m/s^2	Standard value
Load on each wheel	N	1,417.5 N	Step 1
Wheel diameter	D	1.20 m	Configuration-2
Wheel width	b	0.45 m	Configuration-3
Grousers on each wheel	n_{gr}	24	Section 6

Table 4: Soil inputs, from NASA white paper Table 1 [8].

Quantity	Symbol	Value
Sinkage exponent	n	1.0
Cohesive modulus	k_c	1,400 N/m^2
Frictional modulus	k_ϕ	820,000 N/m^3
Cohesion	c	170 N/m^2
Soil weight density	γ	2,470 N/m^3
Shear deformation modulus	K	0.018 m
Internal friction angle	ϕ	33° (Table 1 gives 30–40°; Step 5)
Cohesive bearing factor	N_c	48.09
Density bearing factor	N_γ	33.27
Cohesive deformation modulus	K_c	33.37
Density deformation modulus	K_γ	72.77

3.3 Step 1: Load on Each Wheel

The LLR's fully loaded weight divided by 6 wheels:

$$N = \frac{Mg}{6} = \frac{5250 \times 1.62}{6} = 1{,}417.5 \text{ N} \tag{9}$$

3.4 Step 2: Sinkage

How far the wheel sinks into the regolith under that load [8, Eq. 8]:

$$z = \left[\frac{3N}{(3-n)\,(k_c + b\,k_\phi)\sqrt{D}} \right]^{\frac{2}{2n+1}} \tag{10}$$

The soil stiffness term is $k_c + b\,k_\phi = 1400 + 0.45 \times 820\,000 = 370{,}400$ N/m^2, so

$$z = \left[\frac{3 \times 1417.5}{2 \times 370\,400 \times \sqrt{1.20}} \right]^{2/3} = [0.005240]^{2/3} = 0.0302 \text{ m} \approx 3.0 \text{ cm} \tag{11}$$

3.5 Step 3: Contact Length and Ground Pressure

This angle represents the arc that the wheel contacts regolith with, from first contact to the bottom point of the wheel [8, Eq. 18]:

$$\alpha = \cos^{-1}\left(1 - \frac{2z}{D}\right) = \cos^{-1}\left(1 - \frac{2 \times 0.0302}{1.20}\right) = 18.26^\circ = 0.319 \text{ rad} \tag{12}$$

The contact length is the length of that arc [8, Eq. 24]:

$$l = \frac{D}{2}\,\alpha = 0.60 \times 0.319 = 0.191 \text{ m} \tag{13}$$

This gives a contact area $A = b\,l = 0.45 \times 0.191 = 0.0860$ m^2, and an average ground pressure of

$$p = \frac{N}{A} = \frac{1417.5}{0.0860} = 16.5 \text{ kPa} \tag{14}$$

3.6 Step 4: Compaction Resistance

The force lost compacting the regolith under the wheel [8, Eq. 7]:

$$R_c = (k_c + b\,k_\phi)\,\frac{z^{n+1}}{n+1} = 370\,400 \times \frac{0.0302^2}{2} = 168.9 \text{ N} \tag{15}$$

3.7 Step 5: Soil-Strength Factors

The 'bulldozing' calculation needs four soil-strength factors: N_c, N_γ, K_c and K_γ. These are taken directly from NASA's Table 1 (Table 4). The paper gives the equations behind them, based on Terzaghi's bearing capacity theory [12], but not the friction angle it used, which we need, so it's been reverse-calculated here for later use: [8, Eqs. 12–14, 20–21].

Putting $\phi = 33^\circ$ (0.576 rad, $\tan\phi = 0.6494$, $\cos^2\phi = 0.7034$) into those equations gives:

$$N_q = \frac{e^{(4.712 - 0.576) \times 0.6494}}{2\cos^2(61.5^\circ)} = \frac{e^{2.686}}{2 \times 0.2277} = \frac{14.68}{0.4554} = 32.23 \tag{16}$$

$$N_c = \frac{N_q - 1}{\tan\phi} = \frac{32.23 - 1}{0.6494} = 48.09 \qquad K_c = (N_c - \tan\phi)\cos^2\phi = (48.09 - 0.6494) \times 0.7034 = 33.37 \tag{17}$$

As we can see, a ϕ of 33° matches the NASA values when recalculated.

3.8 Step 6: Bulldozing Resistance

The force lost pushing regolith ahead of the wheel [8, Eqs. 17–19]:

$$R_b = \frac{b\sin(\alpha+\phi)}{2\sin\alpha\cos\phi}\left(2zcK_c + \gamma z^2 K_\gamma\right) + \frac{l_0^3\gamma}{3}\left(\frac{\pi}{2}-\phi\right) + c\,l_0^2\left[1+\tan\left(\frac{\pi}{4}+\frac{\phi}{2}\right)\right] \tag{18}$$

Here l_0 is the length of regolith disturbed ahead of the wheel:

$$l_0 = z\tan^2\left(\frac{\pi}{4}-\frac{\phi}{2}\right) = 0.0302\times\tan^2(28.5^\circ) = 0.0089\text{ m} \tag{19}$$

Working through the terms:

$$\frac{b\sin(\alpha+\phi)}{2\sin\alpha\cos\phi} = 0.667 \tag{20}$$

$$2zcK_c + \gamma z^2 K_\gamma = 506.57\text{ N/m} \tag{21}$$

$$\frac{l_0^3\gamma}{3}\left(\frac{\pi}{2}-\phi\right) = 5.77\times10^{-4}\text{ N} \tag{22}$$

$$c\,l_0^2\left[1+\tan\left(\frac{\pi}{4}+\frac{\phi}{2}\right)\right] = 3.83\times10^{-2}\text{ N} \tag{23}$$

And thus:

$$R_b = 0.667\times506.57 + 0.0006 + 0.0383 = 337.9\text{ N} \tag{24}$$

This application is likely very conservative, as the report applies bulldozing resistance to ALL wheels. In reality, the front two wheels probably reduce bulldozing for the rear four, but the report does not consider this yet.

3.9 Step 7: Rolling Resistance

Adding the two resistances and dividing by the wheel load:

$$C_{rr} = \frac{R_c + R_b}{N} = \frac{168.9+337.9}{1417.5} = 0.357 \approx 0.36 \tag{25}$$

This does not encompass the slope term, as it's assumed to be on level ground for this analysis. However, the slope term is considered later in Step 9. The linear term not covered here, R_r is meant to cover bearing and motor internal friction, but these are already counted elsewhere in drive unit efficiency, so are also ignored.

3.10 Step 8: Grip

The most forward force the soil can enact on a wheel before it shears [8, Eq. 25]:

$$H_{max} = A\,c + N\tan\phi = (0.0860\times170) + (1417.5\times0.6494) = 14.6 + 920.5 = 935.2\text{ N} \tag{26}$$

That is $935.2 \div 1417.5 = 0.66$ as a fraction of wheel load.

The soil only gives this grip when the wheel slips. Slip, s, is how much slower the rover moves than its wheels turn: 0 is pure rolling, and 1 is spinning in place. The grip at a given slip follows the Janosi–Hanamoto relation [10], as given in the NASA paper [8, Eq. 23]:

$$H = H_{max}\left[1 - \frac{K}{s\,l}\left(1 - e^{-sl/K}\right)\right] \tag{27}$$

Table 5: Grip available at different slips, smooth wheel.

Slip s	Share of H_{max}
0.1	0.38
0.5	0.81
1.0	0.91

Counterintuitively, grip builds with increasing slip [8].

3.11 Step 9: Slopes

Next, the 20° slope requirement, Mobility-4, is checked. As a share of the wheel load, the grip needed, μ_{req}, must overcome both the rolling resistance C_{rr} and the downhill pull of the rover's weight, $\tan\theta$:

The rover's weight W splits into two parts, a part pressing into the ground, $W\cos\theta$, and a part pulling it downhill, $W\sin\theta$.

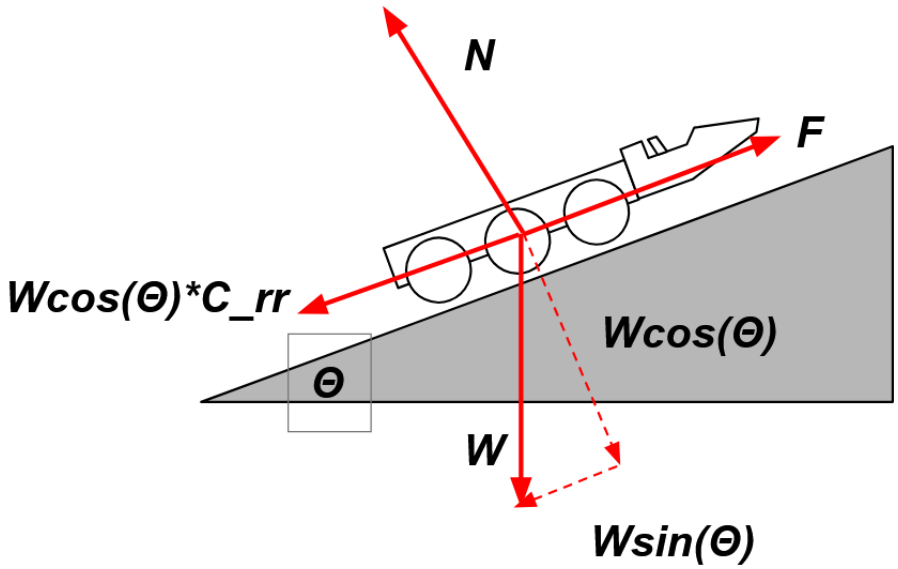


Figure 4: Free body diagram of the LLR in the climbing case, showing relevant forces and components.

$$F = C_{rr}\, W \cos\theta + W \sin\theta \tag{28}$$

$$\mu_{req} = \frac{F}{W\cos\theta} = \frac{C_{rr}\, W\cos\theta + W\sin\theta}{W\cos\theta} = C_{rr} + \tan\theta \tag{29}$$

$$\mu_{req} = C_{rr} + \tan\theta \tag{30}$$

$$\theta = 20^\circ \quad \Rightarrow \quad \tan\theta = 0.364, \qquad \mu_{req} = 0.357 + 0.364 = 0.721 \tag{31}$$

The steepest slope the smooth wheel could hold, even with full grip, is where μ_{req} equals the 0.66 from Step 8:

$$\tan\theta_{max} = 0.660 - 0.357 = 0.302 \quad \Rightarrow \quad \theta_{max} = 16.8^\circ \tag{32}$$

This is below 20°, so the smooth wheel cannot meet the 20° hill climbing goal. In order to fix this, grousers are added in the next steps to provide extra grip.

3.12 Step 10: Adding Grousers

Grousers are extra plates that stand out from the rim and dig into the soil, and are common on rovers, especially lunar rovers. NASA's grouser model [8, Eq. 26] adds a grouser-factor to each of the two parts of the Step 8 grip equation:

$$H_{max} = A\,c\left(1 + \frac{2h_g}{b}N_g\right) + N\tan\phi\left(1 + 0.64\,\frac{h_g}{b}\tan^{-1}\frac{b}{h_g}\right) \tag{33}$$

where h_g is the grouser height and N_g is the number of grousers in contact with the ground [8, Eq. 27]:

$$N_g = \frac{n_{gr}\, l}{\pi D} = \frac{24 \times 0.191}{\pi \times 1.20} = 1.22 \tag{34}$$

3.13 Step 11: Grouser Height for 20°

From Step 9, a 20° slope needs 0.721 of the wheel load:

$$H_{target} = 0.721 \times 1417.5 = 1{,}022.0 \text{ N} \tag{35}$$

Putting in the known values ($A\,c = 14.6$ N and $N\tan\phi = 920.5$ N from Step 8, $N_g = 1.22$ from Step 10, $b = 0.45$ m):

$$14.6\left(1 + \frac{2h_g}{0.45} \times 1.22\right) + 920.5\left(1 + 0.64\,\frac{h_g}{0.45}\tan^{-1}\frac{0.45}{h_g}\right) = 1{,}022.0 \tag{36}$$

h_g appears both outside and inside the $\tan^{-1}$, so the equation cannot be rearranged for h_g directly. Instead, H_{max} was calculated at each centimetre interval of grouser height, to find the shortest grouser that can provide 1,022.0 N.

Example: $h_g = 5$ cm.

$$1 + \frac{2h_g}{b} N_g = 1 + \frac{2 \times 0.05}{0.45} \times 1.22 = 1.2711 \tag{37}$$

$$1 + 0.64 \, \frac{h_g}{b} \tan^{-1} \frac{b}{h_g} = 1 + 0.64 \times 0.1111 \times \tan^{-1}(9) = 1.1038 \tag{38}$$

$$H_{max} = (14.6 \times 1.2711) + (920.5 \times 1.1038) = 18.6 + 1016.1 = 1{,}034.7 \text{ N} \tag{39}$$

Table 6: Full grip against grouser height, 20° target of 1,022.0 N.

h_g	H_{max}
1 cm	956.2 N
2 cm	976.7 N
3 cm	996.6 N
4 cm	1,015.9 N
5 cm	1,034.7 N
6 cm	1,052.9 N

The grip clears the target at 5 cm, so the minimum height lies between 4 and 5 cm. 5 cm is selected to provide a margin, and allow for straightforward design later on.

3.14 Step 12: Slip with Grousers

The full grip from Step 11 is a grip ceiling as it does not account for wheel slippage in the real world. As in Step 8, the soil only gives grip as the wheel slips. The share of full grip it gives at slip s is the Janosi–Hanamoto slip factor, [8, Eq. 23]:

$$f(s) = 1 - \frac{K}{s\,l} \left(1 - e^{-sl/K}\right), \qquad H = f(s)\, H_{max} \tag{40}$$

with $K = 0.018$ m (Table 4) and $l = 0.191$ m (Step 3).

Example: 100% slip. At $s = 1.0$ the wheel spins in place:

$$\frac{s\,l}{K} = \frac{1.0 \times 0.191}{0.018} = 10.61 \tag{41}$$

$$f(1.0) = 1 - \frac{0.018}{0.191} \left(1 - e^{-10.61}\right) = 1 - 0.0942 \times 1.000 = 0.906 \tag{42}$$

$$H = 0.906 \times 1034.7 = 937.4 \text{ N} \tag{43}$$

That is $937.4 \div 1417.5 = 0.66$ of the wheel load.

Table 7: Grip against slip, 5 cm grousers (H_{max} = 1,034.7 N).

Slip s	Slip factor $f(s)$	Grip H	Share of wheel load
10%	0.384	397.3 N	0.28
15%	0.500	517.4 N	0.37
20%	0.585	605.3 N	0.43
50%	0.813	841.2 N	0.59
100%	0.906	937.4 N	0.66

A 20° slope needs 1,022.0 N (Step 11). Even at 100% slip, the 5 cm wheel only produces 937.4 N, 84.6 N short. In this model, the 20° requirement is not met at any real-world slip.

Grouser height for 20° at 100% slip. Working backwards, full grip must be large enough that 0.906 of it reaches 1,022 N.

$$H_{max} = \frac{1022.0}{0.906} = 1{,}128.0 \text{ N} \tag{44}$$

Using the Step 10 equation, 10 cm gives 1,120.0 N and 11 cm gives 1,135.5 N. Therefore, these grousers need to be at least 11 cm tall, just to climb at a very high slip rate, which is unrealistic. 11 cm grousers are far too large to be structurally stable on a 1.2 m wheel, and would likely begin to affect wheel operation due to weight and resistance digging through the soil. As seen in the wheel design study, 5 cm grousers incur a significant mass penalty, and are only rated to slightly above a factor of safety of one. Higher grousers require thicker aluminium, exacerbating the issue.

The model also keeps the smooth wheel's slip behaviour (the same K), so it is likely pessimistic for a grousered wheel, and the real shortfall may actually be smaller. The design therefore keeps 5 cm grousers as the minimum, and the 20° is left to a single-wheel traction test in lunar simulant.

3.15 Future Work

The work shown here is based on old values and methods from the Apollo era. In reality, modern simulation analysis could likely model a far better wheel, with a far better prediction of its performance. In the future, this report recommends that this sort of simulation replace the Bekker analysis method. In addition, any simulation, including the Bekker analysis should be verified independently with real-world physical testing using regolith simulant. For example, the author of this report suspects that the proposed wheel would be able to climb a 20° slope, even if the Bekker analysis disagrees wholeheartedly. Second to that, closing the 20-degree goal can and should be done, so research into potential other soil mechanics models should be conducted.

4 Drive Unit Selection

This section uses the numbers in Section 3 to pick a gearbox and motor combination suitable for the LLR.

4.1 Where the Drive Goes

There were two considered options for motor mounting, inside the wheel hub or on the chassis. Mounting the motor on the chassis makes it the easiest to cool, but a driveshaft would then have to cross both the bogie pivot and the steering axis, resulting in several joints and dust entry points on every wheel.

For these reasons, the in-hub layout was selected. Only power, signal and coolant lines have to cross the steering axis and bogie pivot. A 1.20 m by 0.45 m wheel also has plenty of room inside for a motor and gearbox, and the Apollo LRV used the same layout [24]. The downside of this choice is the thermal consideration. Of course, in a vacuum, there is no air to carry heat away, so all of the motor and gearbox losses have to conduct out through the hub. This is what the pumped fluid loop in Section 5 is for.

4.2 What Each Wheel Must Deliver

The torque each driving wheel needs is its share of the pulling force, times the wheel radius:

$$T = \frac{F}{n}\, r, \qquad F = W\,(C_{rr}\cos\theta + \sin\theta) \tag{45}$$

where n is the number of wheels driving, $W = 8{,}505\,\mathrm{N}$, $C_{rr} = 0.357$ and $r = 0.60\,\mathrm{m}$. Four requirement cases were checked.

Driving at top speed on level ground (Mobility-1), with all six wheels driving:

$$T = \frac{8505 \times 0.357}{6} \times 0.60 = 304\,\mathrm{N\,m} \tag{46}$$

At 4 km/h, with the 15% slip from Section 3.14, the wheels turn at 20.8 rpm.

Climbing the 20° slope (Mobility-4), with all six wheels driving:

$$F = 8505 \times (0.357\cos 20^\circ + \sin 20^\circ) = 5{,}762\,\mathrm{N}, \qquad T = \frac{5762}{6} \times 0.60 = 576\,\mathrm{N\,m} \tag{47}$$

Climbing 15° with one drive unit failed (Mobility-5), so only five wheels are driving:

$$F = 8505 \times (0.357\cos 15^\circ + \sin 15^\circ) = 5{,}136\,\mathrm{N}, \qquad T = \frac{5136}{5} \times 0.60 = 616\,\mathrm{N\,m} \tag{48}$$

The obstacle (Mobility-2) is also considered. The wheel meets the 30.5 cm step at an angle β from straight below the axle:

$$\cos\beta = \frac{r - h}{r} = \frac{0.60 - 0.305}{0.60} = 0.492 \quad \Rightarrow \quad \beta = 60.5^\circ \tag{49}$$

Assuming one wheel has to lift its own load with no help from the others:

$$T = N\, r \sin\beta = 1417.5 \times 0.60 \times \sin 60.5^\circ = 741\,\mathrm{N\,m} \tag{50}$$

A smooth rim could not give this, since it would need a grip of $\mu = \tan\beta = 1.77$ at the edge. With grousers, a grouser hooks the edge and the wheel pivots up over it, so this torque is all that is needed. At 741 N m, this is well inside the gearbox's 1,404 N m repeated peak.

The Mobility-5 case (5 wheels on 15°) sets the torque the drive unit needs, over the 20° slope case (because five wheels have to do the work of six, increasing load).

4.3 Gearbox

A harmonic gearbox was chosen for the LLR. It gives a high gear ratio in a single stage, has no backlash, sits in line with the motor, and was also used to drive the Apollo LRV's wheels [24]. The Harmonic Drive CSG-2A line was compared at a 120:1 ratio against the 616 N m continuous and 741 N m peak requirements [15].

The CSG-40-2A (rated to 382 N m) and the CSG-45-2A (rated to 523 N m) are both below the continuous requirement, so they were ruled out. The CSG-50-2A is rated to 688 N m, with a repeated peak of 1,404 N m, and weighs 3.2 kg. This gives margins of:

$$\frac{688}{616} = 1.12 \text{ continuous}, \qquad \frac{1404}{741} = 1.90 \text{ peak} \tag{51}$$

The gear ratio itself is set by the gearbox's speed limit. Cruising at 4 km/h is something the LLR will do for long periods, so the motor has to stay under the gearbox's average input speed limit. With grease lubrication, this is 3,000 rpm for the CSG-45, 2,500 rpm for the CSG-50 and 2,200 rpm for the CSG-58 [15]. With the wheel turning at 20.8 rpm:

$$20.8 \times 120 = 2{,}496 \text{ rpm at the motor} \tag{52}$$

This is just under the CSG-50's limit, and is the highest standard ratio it allows, since 160:1 would need 3,328 rpm. It also shows why the CSG-58 doesn't work here, as 2,496 rpm is over its 2,200 rpm limit. A higher ratio would let the LLR use a smaller motor, so 120:1 is the best choice for the CSG-50, but it leaves almost no speed margin. This is discussed further in the Future Work section below.

4.4 Motor

The motor has to supply the wheel torque, divided by the gear ratio and the gearbox efficiency. Using the catalogue efficiency of 0.79 at rated torque (2,500 rpm and 40 °C, from the same charts used in Section 5.11):

$$T_{motor} = \frac{616}{120 \times 0.79} = 6.50 \text{ N m continuous}, \qquad \frac{741}{120 \times 0.79} = 7.82 \text{ N m peak} \tag{53}$$

Kollmorgen's frameless motors were considered. A frameless motor is just a rotor and a stator, with no housing or bearings of its own. In the LLR, the wheel hub acts as the housing, the gearbox sits in line with the motor inside it, and the hub bearing carries the load. This removes a housing and a set of bearings from every wheel, and gives the motor a direct path to dump its heat into the hub. Kollmorgen also cites Perseverance heritage, and says its frameless motors can be modified for extreme temperatures, radiation and NASA outgassing limits [26]. It should be noted that this is only a capability Kollmorgen claims, and there is no existing space-qualified version of this motor.

Several Kollmorgen motors were compared against the 6.50 N m continuous requirement [16, 25]. The TBM2G-094-26 (3.17 N m) and the KBM-25X01 (4.90 N m) both fall well short. The TBM2G-115-26 comes close at 6.03 N m, but still fails, and it is the largest motor in its family. The KBM-25X02 gives 8.75 N m continuous and 29.7 N m peak, for margins of 1.35 and 3.80. It has a 110 mm stator, is 93.7 mm long and weighs 3.27 kg. Compared with the TBM2G-115, it is slightly smaller in diameter and longer along the axle, which is the direction a 450 mm wide wheel has room

to spare. The KBM-25X02 was therefore selected. The larger KBM-35X01 (12.6 N m) would also work, but at 140 mm across, it is kept as a fallback.

The catalogue's continuous torque rating assumes the motor is bolted to a large aluminium heatsink in 25 °C air, which a hub in a vacuum doesn't have. For this reason, the margin was checked again assuming the fluid loop keeps the motor's mount at 60 °C. The catalogue gives a motor constant of $K_m = 0.733$ N·m/$\sqrt{\text{W}}$, a thermal resistance of 0.56 °C/W, and a winding limit of 155 °C [16]. The heat from the copper windings is $(T/K_m)^2$, which rises by 0.393% for every degree above 25 °C. At 6.50 N m, the winding settles at about 121 °C:

$$P_{Cu} = \left(\frac{6.50}{0.733}\right)^2 \times 1.38 = 108\,\text{W}, \qquad T_{winding} = 60 + 108 \times 0.56 \approx 121\,^\circ\text{C} \tag{54}$$

This is 34 °C below the limit. Pushed right up to the limit, the motor could hold about 7.77 N m, so the real margin is closer to 1.2 than the catalogue's 1.35, and the motor relies on the coolant loop reaching the hub in order to work at all.

4.5 Selected Drive Unit

Each of the LLR's six wheels is driven by a Kollmorgen KBM-25X02 frameless motor (modified for space) and a Harmonic Drive CSG-50-2A gearbox at 120:1, mounted in line inside the wheel hub. Each unit delivers 616 N m continuous, which is 1.12 times under the gearbox's rating and about 1.2 times under the motor's limit with cooling, and 741 N m peak, well within the gearbox's 1,404 N m repeated peak. At 4 km/h the motor spins at 2,496 rpm. Each drive unit weighs 6.5 kg (3.2 kg gearbox and 3.27 kg motor), or 38.8 kg for all six. This does not yet include the hub housing, bearings and seals, which have not been estimated.

4.6 Future Work

The biggest weakness of this drive unit is its speed margin. At 15% slip, the motor runs at 99.8% of the gearbox's input speed limit, so any extra slip or a change in wheel size would push it over. Dropping to a 100:1 ratio would bring the motor down to 2,080 rpm, or 83% of the limit, but the motor would then need to supply:

$$T_{motor} = \frac{616}{100 \times 0.79} = 7.80\,\text{N m} \tag{55}$$

The catalogue margin for this would be 1.12, but it is slightly over the 7.77 N m the motor can hold with 60 °C coolant, leaving no margin at all. A 100:1 ratio would therefore need either colder coolant at the hub or the larger KBM-35X01 motor.

Additionally, the gearbox's rated torque used above is based on its expected lifespan, so it doesn't actually show how much torque the gearbox can survive. The proper check is the average torque over a real drive cycle, compared with the CSG-50's average torque limit of 1,057 N m. However, a drive cycle for the LLR has not been defined yet, so this check is left for the future.

The 0.79 gearbox efficiency used here comes from the catalogue, for standard grease at 40 °C. A space-rated grease at lunar temperatures has not been tested, and since this efficiency sets both the motor torque and the heat the radiators must reject, it should be measured.

The obstacle torque assumes a single wheel lifts its own load. In reality, the other wheels pushing through the bogie would lower it. However, since Mobility-5 sets the continuous rating, this would not shrink the drive unit.

Finally, the motor and gearbox have not yet been fitted inside the hub in CAD, and the KBM winding option depends on the LLR's bus voltage, which has not been set.

5 Thermal

The rover carries a pumped fluid loop, in theory, much like a car's radiator loop. However, because of the unique nature of the lunar surface, this system must provide two opposite services: heat delivery and removal. During operation, the drive unit motors are artificially cooled to compensate for the lack of natural atmospheric cooling; this is critical, lest they melt or suffer from reduced performance. However, during rest periods, and especially during the lunar night, the sensitive electronics, battery, and motors must be maintained at a minimum temperature to survive, thus drawing on RTG heat. This system is therefore designed to provide two opposite, critical services.

5.1 Architecture Selection

Three ways of keeping the vehicle heated through the lunar night (or while operating in Permanently Shadowed Regions (PSRs)) were considered. Radioisotope Heater Unit (RHU) pellets were advantageous in that this solution would minimize complexity, moving parts, and mass by distributing many individual isotope pellets throughout the machine, attached to sensitive heat-requiring components. Despite this, the risk of lunar contamination by misplacing a pellet in the regolith, increased risk of exposure to astronauts and inability to disable the system during times of excess heat was considered to outweigh any potential benefit, even before considering regulatory 'challenges'.

Therefore, the main competitor to fluid loop heating was electrically distributed heat energy. Its advantage is mechanical simplicity—no moving parts reduces points of failure, maintenance load, and avoids any sort of fluid-in-a-vacuum risk. However, when considering long-term PSR survival, we cannot count on battery electricity to heat the vehicle. Thus, our power source must still be an RTG. RTGs infamously have a low efficiency converting heat energy to electrical potential: NASA's MMRTG produces about $110\,\mathrm{W_e}$ from about $2{,}000\,\mathrm{W_{th}}$ [13]. Additionally, this system is not able to cool the motors, so a separate fluid cooling system would have to be devised, cancelling any sort of mechanical simplicity this choice would provide.

For the above reasons, pumped loop cooling/heating was chosen. The proposed system should be able to operate under two modes, either pumping fluid from the RTG to extremities to provide heat, or pumping fluid from the drive modules to the radiator fins to remove it. In either case, of course, one element needs to be cut off from the system using valves, for example the RTG should not be connected during motor cooling. The MMRTG, in this case, can lose heat through its own radiator fins. In order to minimize crew risk, the MMRTG should be mounted at the rear of the vehicle, be equipped with extra shielding, and real-world dosimetry testing should be done prior to actual use.

5.2 Cold Case: Surviving the Night

Table 8: Cold case inputs.

Input	Value	Source
Night duration	150 h	Environment-2 [5]
LTV drive module heating	10.7 kWh per 150 h, 4 modules	[14] (upper value)
Warm box	$1 \times 1 \times 0.5$ m, 4.0 m^2, held at 0 °C	Judgement
MLI effective emittance, e^*	0.05	[14]
Margin for unmodelled losses	50%	Judgement
MMRTG heat, start of life	~2,000 W_{th}	[13]
Pu-238 half-life	87.7 years	Standard value

5.3 Step 1: External drive units

Taking the LTV's heat loss through four drive units, scaled up to 6:

$$\frac{10\,700\,\text{Wh}}{150\,\text{h} \times 4} = 17.8\,\text{W each}, \qquad 6 \times 17.8 = 107\,\text{W} \tag{56}$$

5.4 Step 2: Warm box (Battery, computers)

Assuming a mid-size warm box, we model the heat lost through Multi-layer Insulation (MLI) to space:

$$Q = e^* \sigma A T^4 = 0.05 \times 5.67 \times 10^{-8} \times 4.0 \times 273^4 = 63\,\text{W} \tag{57}$$

The warm box in this case holds the battery and sensitive internal electronics, for example the 'flight' computer inside an insulated box. For this study, LTV-style MLI was chosen as an example of space-grade insulation. The NASA report suggests using e* = 0.03 for smooth faces, but this report intentionally uses the more conservative 0.05 that NASA used to account for inefficiencies in the real world [14].

5.5 Step 3: Night load

$$Q_{night} = (107 + 63) \times 1.5 = 170 \times 1.5 = 255\,\text{W}_{\text{th}} \tag{58}$$

$$E_{night} = 255 \times 150 = 38.3\,\text{kWh}_{\text{th}} \tag{59}$$

The report added a 50% margin here to account for other losses through the structure, for example. This ensures we are not understating heat loss, even with this simplified model.

5.6 Step 4: RTG at end-of-life

Here, the report checks RTG output at 14 years after launch.

$$Q_{RTG,14} = 2000 \times 0.5^{\,14/87.7} = 2000 \times 0.895 = 1{,}790\,\text{W}_{\text{th}} \tag{60}$$

1,790 W is more than enough to supply the 255 W need during the night. As to the electrical needs of the pump used to drive the fluid loop, the MMRTG fact sheet [13] gives an estimated end-of-design-life power of 72 W at 17 years, which includes three years of fueled storage before launch. This lines up with the 14 years of use checked here.

$$P_{e,14} \approx 72\,\mathrm{W_e} \tag{61}$$

This is lower than the 110 W at start of life, as the MMRTG loses conversion efficiency over its lifespan. However, it should be more than enough to support a fluid pump of the size needed. Below, this is proven using a scale estimate.

5.7 Step 5: Pump power scaling

Mars Pathfinder's pumped loop used a 10 W pump to move 150 W of heat [21]. Scaled to the 255 W night load:

$$P_{pump} \approx 10 \times \frac{255}{150} = 17\,\mathrm{W_e} \tag{62}$$

This uses under a quarter of the RTG's 72 W at 14 years, so the MMRTG should have no problem keeping the LLR warm. In fact, these findings are likely a reason to build a smaller-scale RTG specifically for the LLR, on account of Pu-238 rarity and cost. It should be noted that the pump only needs to run under RTG power during the cold case. The hot case would likely require more flow rate, but can operate under battery power, so the limitation is not in place.

5.8 Hot Case: Driving

For the 'Hot case', the LLR is considered to be driving on level ground, at maximum speed (4 km/h) and load (3,500 kg) with the sun at its highest (7.5°). Every watt lost to inefficiency in the drivetrain is considered to be needed to remove by the fluid loop.

Table 9: Hot case inputs.

Input	Value	Source
Loaded mass	5,250 kg	Payload-1, Payload-2
Rolling resistance, C_{rr}	0.357	Section 3.9
Slip, level ground	15%	Section 3.14
Speed	4 km/h (1.111 m/s)	Mobility-1
Wheel radius	0.60 m	Configuration-2
Gearbox	CSG-50-2A, 120:1, 688 N m rated	[15]
Motor	KBM-25X02	[16]
Controller	AKD-x00607	[17]
Coolant temperature at cold plates	60 °C	Judgement
Radiator emittance, ε	0.92 (Z-93 white paint)	[22]
Radiator absorptance, α	0.17 (Z-93 white paint)	[22]
Solar flux	1,414 W/m^2	Standard value (perihelion)

5.9 Step 1: Power at the wheels

$$F = mgC_{rr} = 5250 \times 1.62 \times 0.357 = 3{,}036\,\text{N} \tag{63}$$

$$P_{ground} = Fv = 3036 \times 1.111 = 3{,}374\,\text{W} \tag{64}$$

$$P_{wheels} = \frac{P_{ground}}{1 - s} = \frac{3374}{0.85} = 3{,}969\,\text{W} \tag{65}$$

The slip value here accounts for the wheels 'slipping' on regolith, spinning about 18% faster than they would with perfect contact at perfect friction.

5.10 Step 2: Operating point of wheels

Angular speed and torque of each wheel is found here, using the numbers above.

$$\omega_{wheel} = \frac{1.111}{0.60 \times 0.85} = 2.18\,\text{rad/s} = 20.8\text{ rpm} \quad \Rightarrow \quad 20.8 \times 120 \approx 2{,}500\text{ rpm at the motor} \tag{66}$$

$$T_{wheel} = \frac{3036}{6} \times 0.60 = 304\,\text{N m} = 0.44\text{ of rated gearbox torque of 688 Nm} \tag{67}$$

5.11 Step 3: Chaining efficiencies of drivetrain components

Table 10: Drivetrain efficiency at 2,500 rpm and 0.44 rated torque.

Stage	Efficiency	Source
Gearbox	$0.79 \times 0.87 = 0.687$	[15], pp. 38–39
Motor	0.90	[16], p. 26
Controller	0.95	[17], §7.6
Chain	**0.587**	

The gearbox has a 0.79 efficiency, read from the catalogue chart at 2,500 rpm at the maximum listed temperature of 40°C, and multiplied by the 0.87 factor from partial load at 0.44 rated torque. These numbers were pulled from the charts in Figure 5.

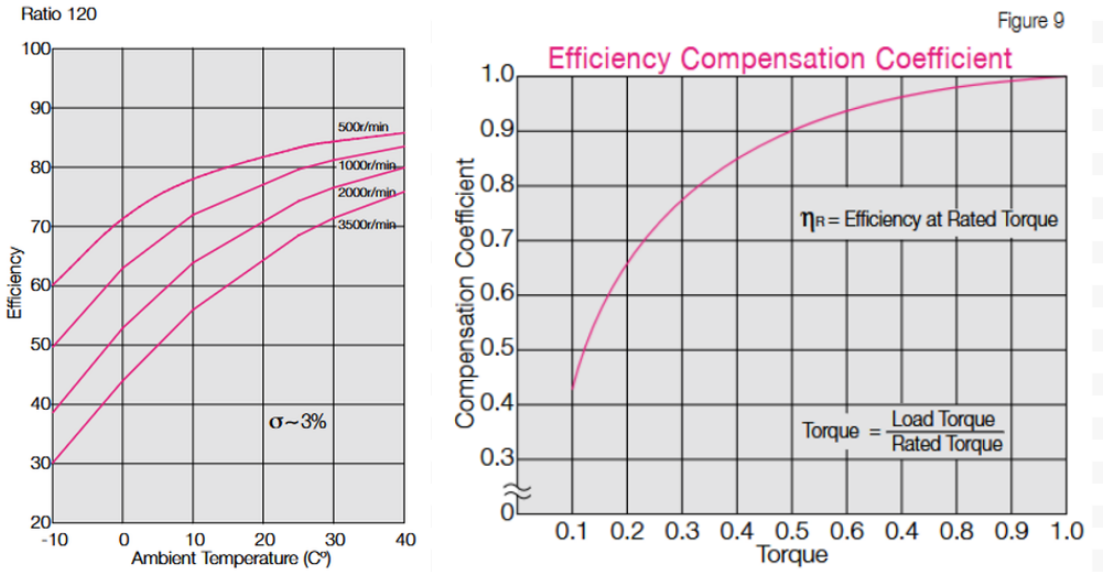


Figure 5: Efficiency charts of the gearbox at the operating points assumed in this report.

5.12 Step 4: Calculating heat entering the loop

The drivetrain delivers 58.7% of its input power to the wheels and the other 41.3% is lost as heat.

$$Q_{hot} = P_{wheels}\left(\frac{1}{\eta} - 1\right) = 3969 \times \left(\frac{1}{0.587} - 1\right) = 3969 \times 0.704 = 2{,}790\,\text{W} \tag{68}$$

Gearboxes have by far the lowest efficiency, and thus produce the most heat. Therefore, the loop design must be careful to optimize heat removal from the gearboxes in particular.

5.13 Step 5: Sizing a radiator

All Artemis III candidate regions are within 6° of the pole [18], and the Moon's axis tilts 1.54°. For this report, this has been simplified to a worst-case scenario of the sun at 7.5 degrees above the horizon. Interestingly, due to the low-sun angle, radiator fins should not be mounted vertically, as they would act like solar absorbers, opposite to intended effect. For this reason, radiator fins are mounted flat, horizontally off the edge of the LLR, as seen in the CAD drawings, Figure 1. The radiator is painted with Z-93 white paint ($\alpha = 0.17$, $\varepsilon = 0.92$ [22]), the same type of coating as VIPER's radiators (Z93C55) [23].

$$\text{Heat out} = \varepsilon\sigma T^4 = 0.92 \times 5.67 \times 10^{-8} \times 333^4 = 641\,\text{W/m}^2 \tag{69}$$

$$\text{Sun in} = \alpha\, S \sin 7.5° = \alpha \times 1414 \times 0.131 \tag{70}$$

Table 11: Radiator area for Z-93 white paint.

Case	α	Sun in	Net out	Area (+30%)
Design case	**0.17**	31 W/m^2	610 W/m^2	5.9 m^2

$$A = \frac{2790}{610} \times 1.3 = 4.57 \times 1.3 = 5.9\,\mathrm{m}^2 \tag{71}$$

In this case, white Z-93 was chosen as an example of white, space-grade paint. The radiators on the LLR are planned to hang off the edge of the vehicle, and be exposed to flying debris and dust as the rover drives. For this reason, fancier materials like Quartz-mirror optical solar reflectors (OSRs) were rejected. In addition, the NASA VIPER rover used similar white-paint radiators [23].

5.14 Results

Table 12: Thermal results.

Quantity	Value	Status
Night heat load	$255\,\mathrm{W_{th}}$	Includes 50% margin
RTG heat at 14 years	$1{,}790\,\mathrm{W_{th}}$	Meets, 7.0× the load
RTG electrical output at 14 years	$72\,\mathrm{W_e}$	Fact sheet end of life; pump needs $\sim$17 $\mathrm{W_e}$
Heat while driving	2,790 W	
Radiator area needed	$5.9\,\mathrm{m}^2$	Fit not yet shown (Envelope-1)

5.15 Future Work

In effect, this section represents back-of-the-napkin math with numbers pulled from published data. A far more accurate analysis would involve first sizing a battery based on the mobility analysis, then designing a warm box to encompass the battery and electronics. This model of a warm box could then be used in a computer-analysis of thermal loss, providing a far more accurate estimation of heat needs. In addition, as said by the report, the actual cold-case heat and energy needs fall far below what the MMRTG can provide, even at the end of its life. A custom RTG unit could be explored to reduce the size, cost and hazard associated with a full MMRTG. As it stands the actual hazard to the crew of MMRTG exposure is untouched, and assumes crew-protective shielding is possible. However, to avoid the mass and safety penalty, a custom or reduced RTG would be a better solution. In fact, the MMRTG has an inherently modular design, so the solution could simply be removing fuel canisters to a minimum amount.

6 Wheel Design Study

The wheels are perhaps the most critical parts of a transport rover, and the ones subject to the most wear. Combined with the harsh conditions and the little design precedent available, this made the wheel one of the more difficult parts of the LLR to design.

Standard rubber or pneumatic tyres have historically been rejected because of UV and temperature degradation, especially over multiple lunar night cycles [7].

The Apollo Lunar Roving Vehicle (LRV) is the closest example in size and mass on the lunar surface, and so was the first source of inspiration. The wire-mesh construction of its wheels, with chevron tread, gave a flexible, conforming wheel that crossed the lunar surface smoothly. However,

later studies for larger, heavier vehicles found that wire-mesh construction did not scale easily to higher loads: manufacturing became increasingly difficult, and the wheel fared worse under higher stress [7].

Other concepts were therefore considered. The Perseverance rover and its predecessor, Curiosity, both use stronger wheel constructions for long-term operation. The concept applied here is titanium inner flexures supporting a thin aluminium shell, reinforced with grousers for traction and durability. Curiosity has since suffered significant wheel damage [19]. Despite this, these wheels give a basis for a new high-load lunar wheel.

6.1 Problem Definition

The worst-case load used in the simulations below is 3,402 N: a fully loaded rover resting on only three wheels, with a 1.2 load factor, applied to a single flexure on the bottom face where it attaches to the supporting ring:

$$F = \frac{Mg \times 1.2}{3} = \frac{5250 \times 1.62 \times 1.2}{3} = 3{,}402\,\text{N} \tag{72}$$

This is an intentionally conservative and demanding estimate. The wheel flexures are a critical component of the wheel, itself a critical component of the LLR. A wheel failure during a mission could slow or stop the rover, potentially putting astronaut lives at risk, which is not acceptable for a crewed vehicle.

6.2 Design

The final wheel design consists of titanium wheel flexures mounted between an aluminium axle connection and an outer hoop (spine), giving a strong, shock-absorbing design. The design also uses individual wheel grousers, drawing on previous Canadian lunar wheel design [20].

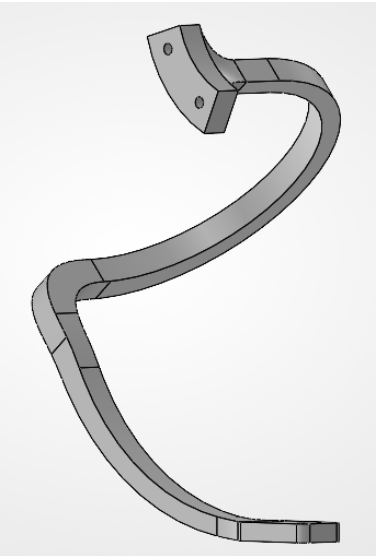


Figure 6: Final design of the titanium wheel flexure.

Figure 7 shows the FEA of the initial wheel flexure design. This initial design fails in the midsection under the test load.

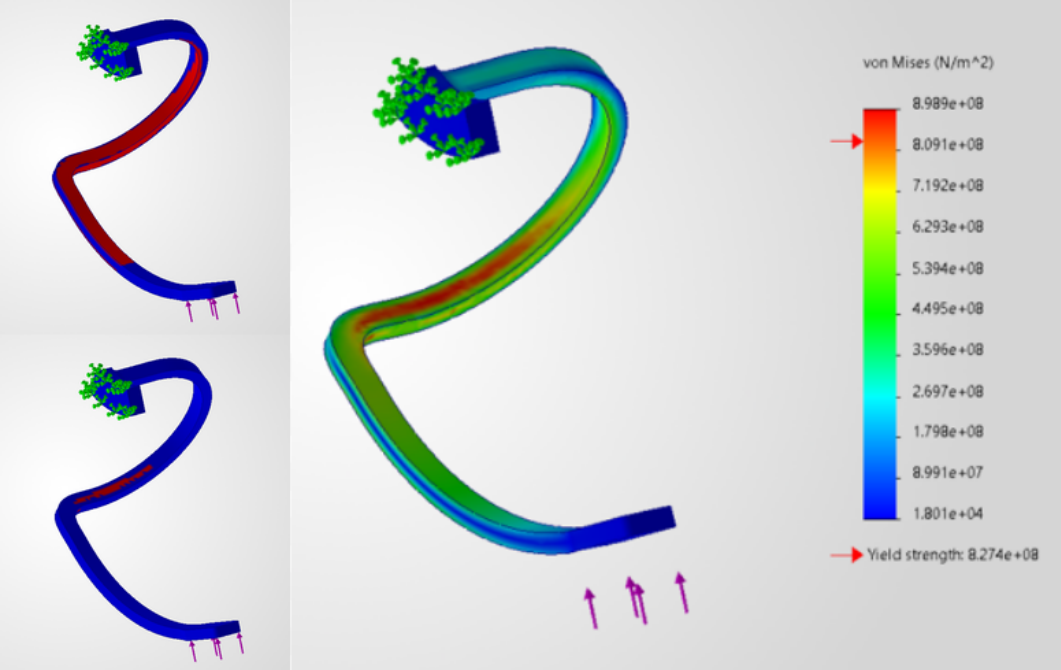


Figure 7: Initial flexure FEA. Clockwise from top left: areas in red are below FoS 2; von Mises stress; areas in red are below FoS 1, indicating failure.

After redesign, thickening the most vulnerable area gives the result in Figure 8. M12 placeholder fastener holes have also been added, and the axle attachment point has been centred for easier fastener placement.

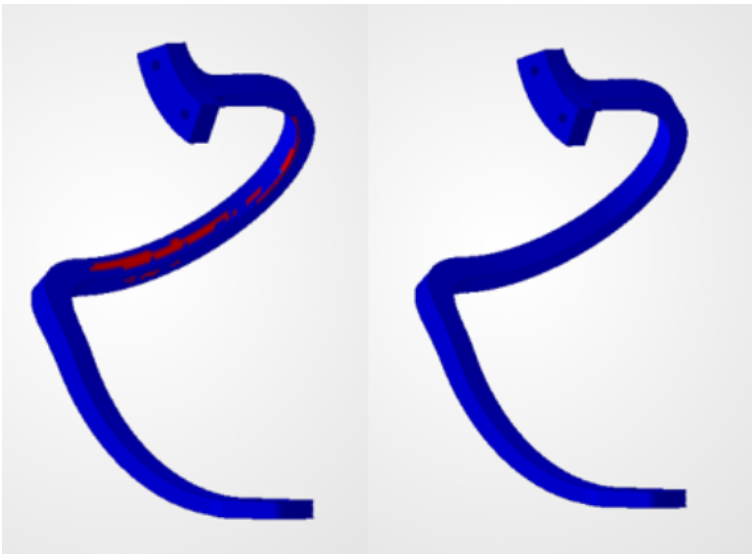


Figure 8: Final flexure FEA. Areas in red indicate FoS below 2 (left) and 1.5 (right).

The part is now rated to a factor of safety of 1.68 in the worst-case load case above, and has a mass of 5.324 kg of Ti-6Al-4V titanium alloy, a standard aerospace alloy. Titanium was selected for its higher strength-to-weight ratio compared to aluminium, allowing a lighter design. Though expensive to manufacture, titanium was also used for similar components on the Curiosity and Perseverance rovers, because of the critical nature of the part and the small quantity required.

After the flexures, the spine (Figure 9) and the axle fixture point were designed. Their less demanding loads allowed aluminium to be used, cutting manufacturing and material cost.

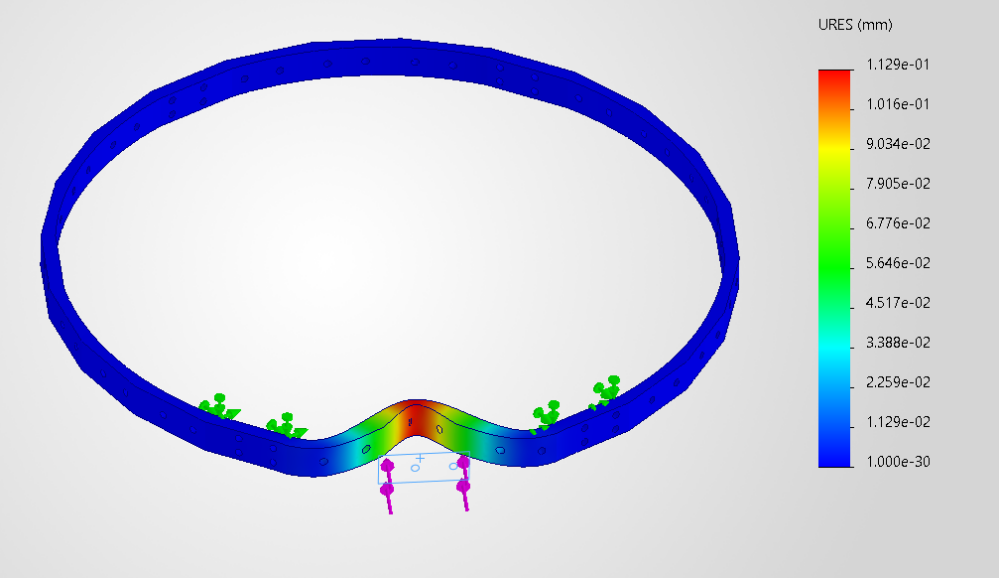


Figure 9: The outer spine in its worst-case load simulation: the full force on one grouser exactly between two flexures. The displacement is shown greatly exaggerated; the ring actually moves by about 0.1 mm.

The spine is a circular ring with 24 flat faces on the outside to mount grouser plates. The grouser plates are sacrificial by design, and this modular approach lets astronauts swap damaged plates over the life of the rover.

The last part of the wheel to be designed was the grousers. A thin aluminium skin was originally planned, similar to the Perseverance and Curiosity wheels. However, this has been one of the failure points of those wheels: repeated wear has broken through the thin aluminium skin and degraded wheel performance. This would not suit the higher speeds, loads, and abrasive unweathered lunar soil the LLR would meet.

Instead, a Canadian design approach was used: individual aluminium grousers, based on the SPRP and Lunar Rover Platform and Drivetrain Prototype (LRPDP) rover concepts developed by the Canadian Space Agency (Figure 10). This construction gives higher durability, easy replacement and design flexibility, at the cost of higher mass from its more complex construction.

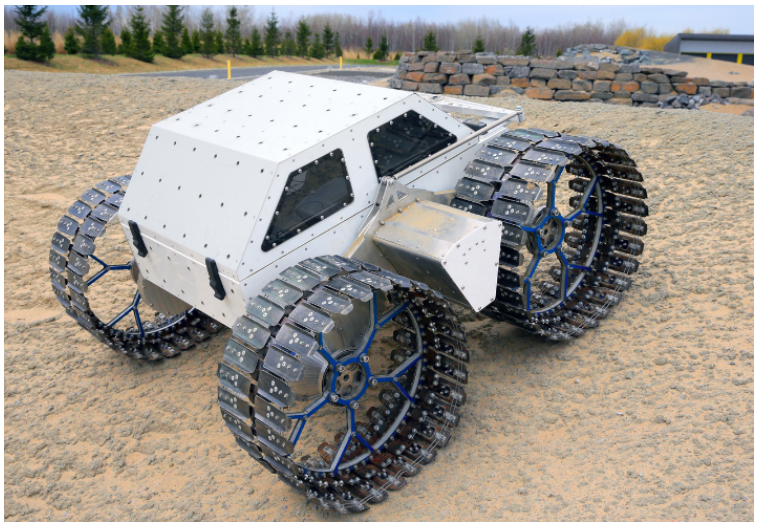


Figure 10: The CSA's Small Planetary Rover Platform (SPRP), with grousered frame wheels similar to the LLR's design. Image: Canadian Space Agency [20].

The grouser design is preliminary, using a bent sheet-metal blank with bolt holes (Figure 11). This allows very cheap construction compared with the thin-skin designs used by NASA. When considering the full-load case of 3,402 N placed directly on the edge of one half-lip of the grouser (as could be the case should the rover need to directly overcome the rock), the grouser handles it with minor deformation and a factor-of-safety of 1.12 (outlined in Figure 12). This is considerably lower than the 1.68 of the titanium flexure. This is acceptable in the design because they are designed to withstand abuse to protect other components and be replaced. Manufacturing these with a higher FoS is possible, but would incur a higher mass, especially when considering the quantity required. For this reason, they have been left with a lower FoS, considering the relative cheapness and replaceability compared to the titanium flexures.

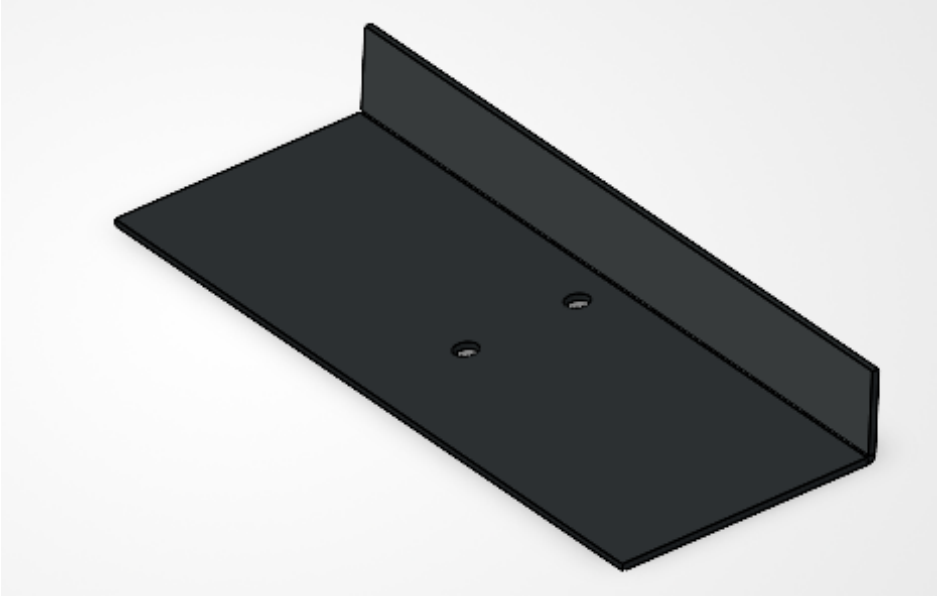


Figure 11: Sheet-metal aluminium grouser proposal, cut from 4.0 mm sheet 7075-T6 Aluminium, with bent edges for grip. The grousers stand 5 cm tall, the minimum set in Section 3.

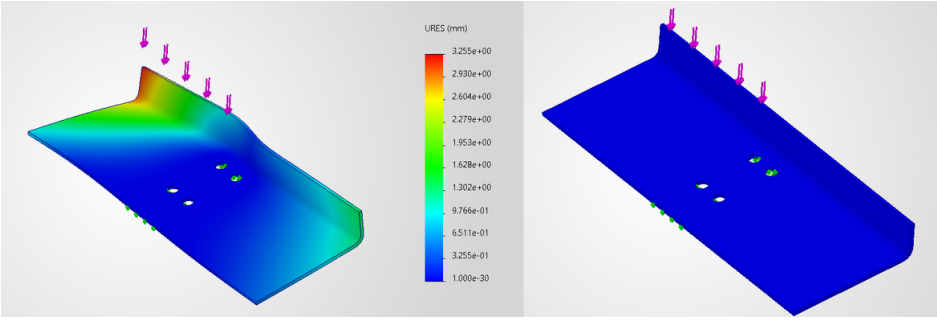


Figure 12: Grouser performance under $1.2 \times 1/3$ LLR weight at max load, applied to one half-edge of the lip. Both displacement (left) and areas under FoS 1 (right, note the lack of red, indicating the entire part can withstand the load safely) are shown.

The final design is shown in Figure 13. Extensive real-world testing will be needed, even after further simulation. However, this concept puts together the titanium flexures of the Mars rovers and the replaceable grousers of the CSA's rover concepts, for a wheel larger and more heavily loaded than any lunar wheel flown so far. The final CAD model is 75.1 kg, before fasteners. Including a 20% fastener budget, each wheel is an estimated 90 kg, and all six wheels at 540 kg. This represents a significant portion of the 1,750 kg dry mass target.

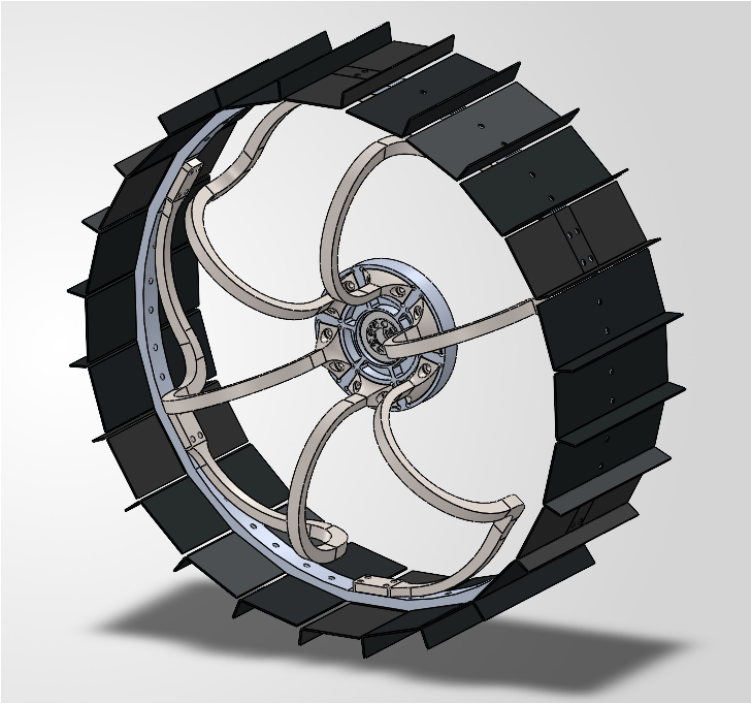


Figure 13: The LLR wheel concept, with Mars rover-style titanium flexures and CSA-style replaceable grousers.

6.3 Future Work

The model of a wheel here is provided to be the best-guess at what a wheel would look like for this mission, taking into account both historical challenges and the needs of this current mission. However, it's likely not the final design, and real-world testing should be employed to verify the numbers and findings reported in this section. In the future, a scale model of the wheel should be mounted on an apparatus that can apply the estimated force of the LLR, and roll it across lunar regolith simulant landscapes and slopes. This testing method would verify the LLR's performance on the Moon, instead of just an on-paper proof-of-concept like what was provided here. Finally, the 540 kg estimated total wheels mass represents a significant chunk of the targeted 1,750 kg dry mass target. The design may need to be reconsidered to allow for lighter weight or fewer parts.

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A Decision Register

This register lists every requirement and decision that the report uses or checks. Each one is either an input set at the start of the project, or a decision that came out of one of the studies. Sources are the CSA, NASA, the Apollo LRV program, industry reporting, this project's own studies (Trade), values calculated from other entries (Derived), or the author's own judgement where no outside source exists (Judgement). Payload-2 is a target; every other entry is fixed.

ID	Requirement	Source	Status
Payload and crew			
Payload-1	The LLR shall carry a 3,500 kg payload, crew included.	Industry	Not assessed (Payload-2)
Payload-2	The LLR shall have a dry mass of no more than 1,750 kg.	Judgement	Target, not assessed
Payload-3	The LLR shall carry 2 suited crew in an unpressurised crew station with restraints.	Judgement	By design (Sec. 1)
Configuration and wheels			
Configuration-1	The LLR shall have 6 driven wheels on 3 axles.	Trade	By design
Configuration-2	The wheels shall be 1.20 m in diameter.	Derived	By design
Configuration-3	The wheels shall be 0.45 m wide.	Derived	Met (Sec. 3)
Configuration-4	The track shall be 2.40 m.	Trade	By design

ID	Requirement	Source	Status
Configuration-5	The wheelbase shall be 3.50 m, with even 1.75 m axle spacing.	Trade	By design
Configuration-6	The wheels shall be non-pneumatic, with a rigid spine ring joined to the hub by titanium flexures.	Trade, Apollo	Met (Sec. 6)
Configuration-7	Each wheel flexure shall carry 3,402 N with a factor of safety of at least 1.5 on yield.	Judgement	Met, FoS 1.68 (Sec. 6)
Mobility			
Mobility-1	The LLR shall reach 4 km/h on level ground at full load.	Judgement	Met (Sec. 4)
Mobility-2	The LLR shall climb a 30.5 cm obstacle.	Apollo	Met by analysis (Sec. 4); test to confirm
Mobility-3	The LLR shall have at least 40 cm of ground clearance.	Derived	By design
Mobility-4	The LLR shall climb and descend a 20° slope at full load, at 1 km/h or faster.	NASA LTV	Not met (Sec. 3.14)
Mobility-5	The LLR shall climb a 15° slope at full load with any one drive unit failed.	Trade	Not met (Sec. 2.3)
Environment			
Environment-1	The LLR shall traverse a PSR for at least 2 h with crew aboard, in temperatures down to 25 K.	NASA LTV	Not assessed (Sec. 2.3)
Environment-2	The LLR shall survive 150 h dormant and unoccupied.	NASA LTV	Met for heat (Sec. 5.2)
Environment-3	The LLR shall support a crewed sortie of up to 8 h.	NASA LTV	Not assessed (Sec. 2.3)
Environment-4	The LLR shall have a design life of at least 10 years, with degrading parts checked at 14 years.	CSA	Met for RTG heat (Sec. 5.6)
Envelope and stowage			
Envelope-1	The LLR shall be no more than 2.90 m wide when deployed.	Derived	Not assessed
Envelope-2	The LLR shall stow inside a 5.75 m diameter cylinder.	Derived	Met in plan view (Sec. 1)
Power			
Power-1	The LLR shall be powered by one MMRTG (about 2,000 Wth and 110 We at start of life).	NASA	Input; 72 We at 14 years (Sec. 5.6)
Power-2	The battery shall be sized for one 8 h crewed sortie.	Derived	Not assessed
Power-3	The dormant electrical load shall not exceed the RTG's electrical output at end of life.	Derived	Met for pump, 17 We vs 72 We (Sec. 5.7)
Thermal control			

ID	Requirement	Source	Status
Thermal-1	The LLR shall use a pumped fluid loop to move RTG heat to cold parts when dormant, and drive unit heat to the radiators when driving.	Trade	Met (Sec. 5)
Thermal-2	The radiators shall face the sky, with no vertical radiator surfaces.	Derived	By design (Sec. 5.13)
Operations			
Operations-1	The LLR shall go no more than 2.0 km into a PSR.	NASA, Judgement	Not assessed
Operations-2	The LLR shall only enter a PSR with enough energy to finish its task and drive out.	Derived	Not assessed
Operations-3	The LLR shall be operable crewed, teleoperated from Earth, and in supervised autonomy.	CSA	Not assessed (Sec. 1)